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EURES Report on labour shortages and surpluses
2024

Sector analysis on warehousing and support
activities for transportation

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Executive summary

- This sector analysis on warehousing and support activities for transportation (NACE Class H52) accompanies the EURES Report on labour shortages and surpluses 2024. This analysis focuses on the main determinants and causes of labour market imbalances in the sector.
- In 2024, the sector was made up of approximately 165 000 enterprises directly employing around 3.2 million individuals. At the EU level the warehousing and support activities for transportation sector, there is a surplus of freight handlers and shortage of mechanics and repairers.
- The occupations of aircraft engine and motor vehicle mechanics and repairers present the most severe labour shortages. This is followed by a shortage in air traffic controllers. Although most countries indicate a surplus of general freight handlers, stakeholders reveal the sub-group of airport ground-handling operations frequently encounter labour supply constraints.
- Privatisation and increasing competition in the water, aviation, and railway sectors have led to lower working conditions. At the same time, digitalisation and automation are changing employment trends, creating more demand for medium- and high-skilled workers. As a result, there is a growing shortage of skilled workers. Automation is also beginning to replace jobs across all sectors, particularly affecting lower-skilled roles.
- The COVID-19 pandemic led to widespread layoffs in transportation support services, with many workers leaving for other sectors, making rehiring challenging. The pause in recruitment and training during this period contributed to a long-term shortage of skilled workers, particularly in aviation and maritime support services, as transport activity returned to pre-pandemic levels. In the aftermath of the COVID-19 pandemic, many workers left the warehousing sector to return to other sectors, leading to some shortages.
- The sector struggles to replace retiring workers, with additional challenges related to the health and capability needs of an ageing workforce. Low awareness of career opportunities and working conditions deter younger workers, while gender stereotypes, limited career progression, and inadequate workplace facilities contribute to the underrepresentation of women, particularly in technical roles.
- There are more migrant workers and EU nationals working in other EU countries in the warehousing and support activities for transport compared to other transport sectors. The warehousing sector has one of the highest numbers of posted workers in the EU, with an increasing share of third-country nationals.
- Harsh working conditions, low wages, job insecurity, and limited career progression contribute to high turnover and labour shortages. Temporary contracts, subcontracting, and the rise of automation further discourage new recruits, particularly in aviation ground handling and maritime industries. These factors, along with poor job stability and unattractive long-term prospects, make it difficult to retain and attract workers.
- Labour market imbalances in the sector are driven by a growing skills gap, as digitalisation and increasing complexity require advanced technical skills and higher education. Insufficient training infrastructure, slow adaptation of educational programmes, and a lack of in-house training hinder skills development.
- Recruitment practices relying heavily on subcontracting and temporary contracts create fragmented workforces with poor conditions, low job stability, and high turnover. Seasonal demand in sectors like ports and airports limits permanent hiring, while wages and lack of training further reduce attractiveness and retention.
- To tackle labour market imbalances, companies and Member States are implementing measures ranging from strengthening training, improving sector attractiveness, promoting gender equality and diversity and automation. These take a wide range of forms, including legislation promoting safety procedures in the ground handling sector, research and innovation, diversity and inclusion initiatives, community engagement, competitive salaries and improving the physical work environment.

1.0 Introduction

This sector analysis accompanies the EURES Report on labour shortages and surpluses 2024, with this year's edition focusing on the transport sector. The purpose of this analysis is to investigate labour market imbalances in the warehousing and support activities for transport sector, with a particular focus on understanding their determinants and causes.

Based on the NACE classification (Class H52), the sector encompasses:

- Warehouse and storage: Operation of storage and warehouse facilities for all kinds of goods. It involves stationary operations focused on logistics support.
- Support activities for transportation that do not involve the actual movement of goods and people and include enabling tasks like managing the logistic chains or maintaining transport hubs. These cover service activities incidental for transportation via:
 - Land: operation of parts of the transport infrastructure or activities related to handling freight immediately before or after transport or between transport segments (e.g. railway stations, bus stations, goods handling stations, railroad infrastructure)
 - Water: operation of terminal facilities such as harbours and piers, management of waterway locks, navigation, pilotage, berthing activities, lighterage, salvage operations, and lighthouse services
 - Air: operation of terminal facilities such as airports, air-traffic control activities, and ground services at airfields

An overview of the sector is provided in Chapter 2. The labour market imbalances in this sector were analysed in relation to the following macro topics:

- Influence of sectoral trends on labour market imbalances (Chapter 3)
- Recent global crises: implications for the labour market (Chapter 4)
- Workforce demographic and related issues (Chapter 5)
- Labour migration and mobility (Chapter 6)
- Working conditions and impacts on industry's reputation and the role of social dialogue (Chapter 7)
- Skills and qualifications gaps (Chapter 8)
- Recruitment practices to fill labour and skill gaps (Chapter 9)
- Measures to tackle labour market imbalances (Chapter 10)

The study draws on a comprehensive literature review conducted at both EU and national levels, supplemented by insights from a focus group (held online on 27 November 2024) comprising key stakeholders, including representatives from trade unions and industry operators. Further data were provided by EURES National Coordination Offices (NCOs), who identified occupations within the sector experiencing shortages and surpluses. Additionally, Eurostat data were used to highlight key indicators related to workforce profiles.

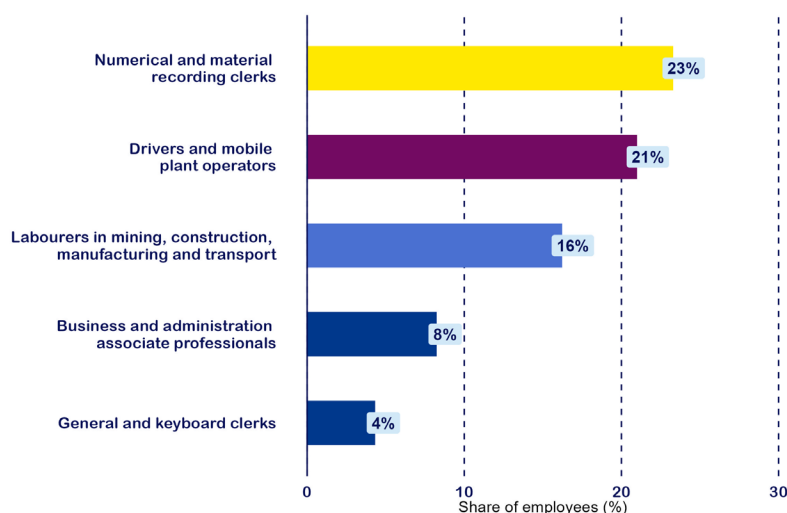
2.0 Overview of the warehousing and support activities for transportation

As of 2022, the warehousing and support activities for transportation accounted for around 27% of the employment in the transportation and storage sector in the EU (European Commission, 2024b). At the end of 2024, the sector directly employed around 3.2 million workers (Eurostat, 2024b)

With around 165 000 enterprises, the sector is highly concentrated. Large enterprises with 250 or more employees make up only 1% of businesses but employ 55% of the workforce. In contrast, micro enterprises with 0-9 employees (85%) and small enterprises with 10-49 employees (12%) dominate in number but contribute less to total employment (25% combined) (Eurostat, 2024c).

Figure 1 shows numerical and material recording clerks (23%) and drivers (21%) make up the largest occupations in the sector's workforce. Labourers (16%) play a significant role in manual tasks, while business and administration professionals (8%) support operations.

Figure 1 - Share of employment represented by the five most common broad occupations in the warehousing and support activities for transportation sector, 2023



Source: Eurostat (2024). Labour Force Survey special extractions.

Table 1 gives an overview of the countries reporting shortages and surpluses for selected warehousing and support activities for transportation occupations. The greatest reported shortage is for motor vehicle mechanics and repairers, with sixteen countries reporting shortages. The second most severe is aircraft mechanics and repairers with nine countries experiencing such shortage. In terms of surplus, 14 countries have a surplus of freight handler employees (land, water, air combined).

The data presented in this table was collected through a survey distributed to the EURES National Coordination Offices (NCOs) of the 27 Member States of the European Union (EU), Iceland, Norway, Switzerland, and Liechtenstein. Through this survey, each NCO provided information on the match or mismatch status for various occupations. Some of these data should be interpreted with caution. In the case of surplus of aircraft engine mechanics and repairers in Denmark, trade union stakeholders consulted in the focus groups emphasized a significant demand for more qualified aircraft technicians. Similarly, for Austria, trade union stakeholders consulted in the focus groups reported that air traffic controllers and safety electronic technicians are working overtime due to shortages.

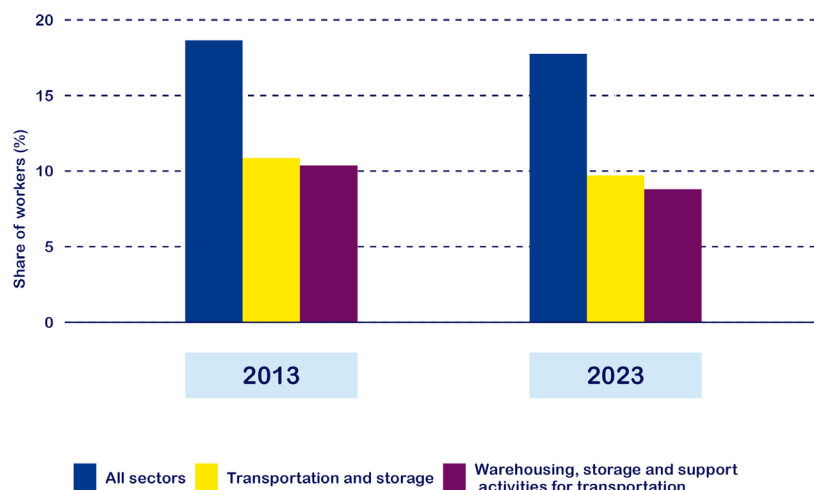
Table 1 - Countries reporting labour shortages and surpluses for selected warehousing and support activities for transportation sector occupations, 2024

Occupation	Countries reporting shortages	Countries reporting surpluses
Aircraft engine mechanics and repairers	9 countries (BE, DE, HU, LT, LU, MT, AT, RO, SK)	3 countries (CZ, DK, FI)
Motor vehicle mechanics and repairers	16 countries (BE, BG, DE, DK, EE, FR, HR, IT, LT, MT, NL, NO, PL, PT, SE, SI)	5 countries (ES, FI, PT, RO, SK)
Bicycle and related repairers	3 countries (DE, DK, NL)	4 countries (ES, FI, HU, AT)
Supply, distribution and related managers	6 countries (BE, FR, NL, MT, RO, SK)	7 countries (BE, BG, ES, LT, LV, AT, PT)
Freight handlers	5 countries (CY, HU, MT, NL, SK)	14 countries (BE, BG, DE, DK, EL, FI, LT, LU, LV, NO, AT, PT, RO, SK)
Air traffic safety electronics technicians	5 countries (DE, EL, ES, FI, RO)	5 countries (DK, HU, LV, MT, AT)
Air traffic controllers	5 countries (BE, ES, LV, MT, SK)	3 countries (DK, AT, RO)

Source: NCO data. NCOs may report both a shortage and a surplus for a given occupation due to differences at regional level.

Part-time work in the warehousing, storage and support activities for transportation is slightly lower than for the transportation and storage sector and significantly lower than for all sectors (Figure 2). The prevalence of workers with part-time contracts has slightly decreased in the past decade.

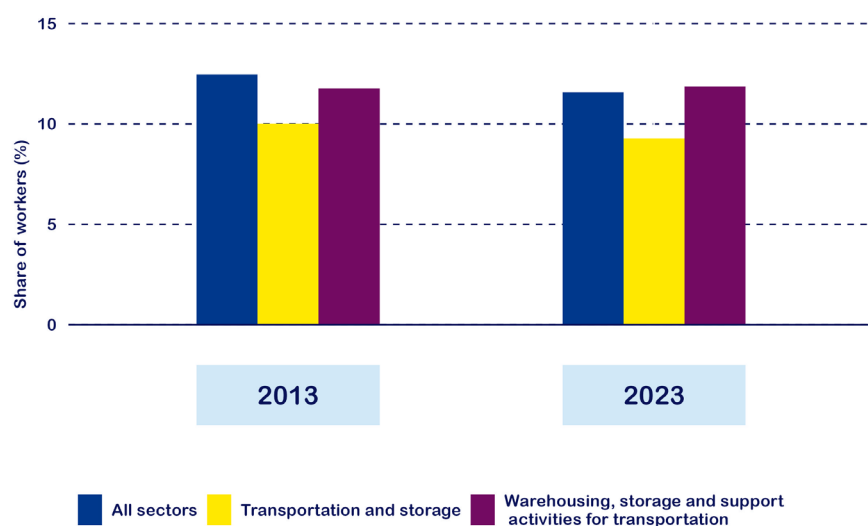
Figure 2 - Part-time employment in the warehousing and support activities for transportation sector, the transportation and storage sector and the overall economy in the EU-27, 2013 and 2023



Source: Eurostat (2024). Labour Force Survey special extractions.

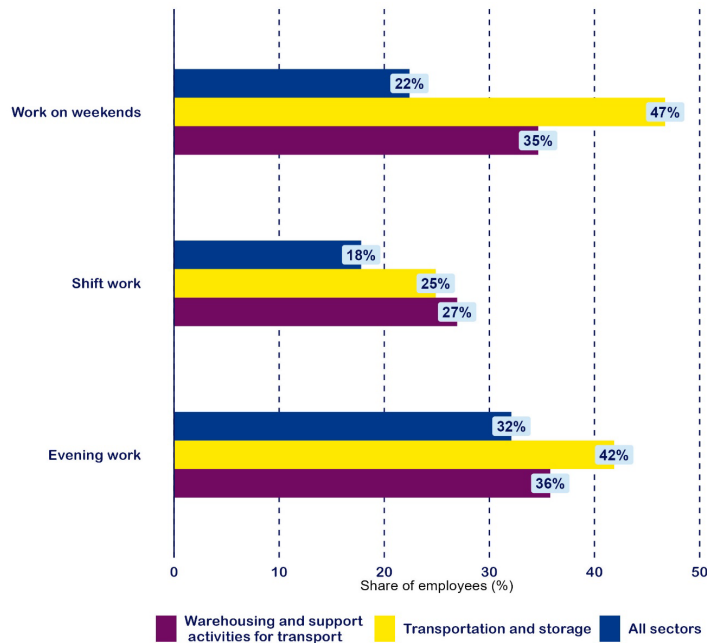
The nature of warehousing and support activities for transportation leads to non-standard work schedules. Indeed, Figure 3 shows this sector has on average more work on weekends, shift work and evening work than all sectors. The sector has more shift work than the transport sector, but less work on weekends and evening work than the transport sector average (Figure 4).

Figure 3 - Temporary employment in the warehousing and support activities for transportation sector, the transportation and storage sector and the overall economy in the EU-27, 2013 and 2023



Source: Eurostat (2024). Labour Force Survey special extractions. .

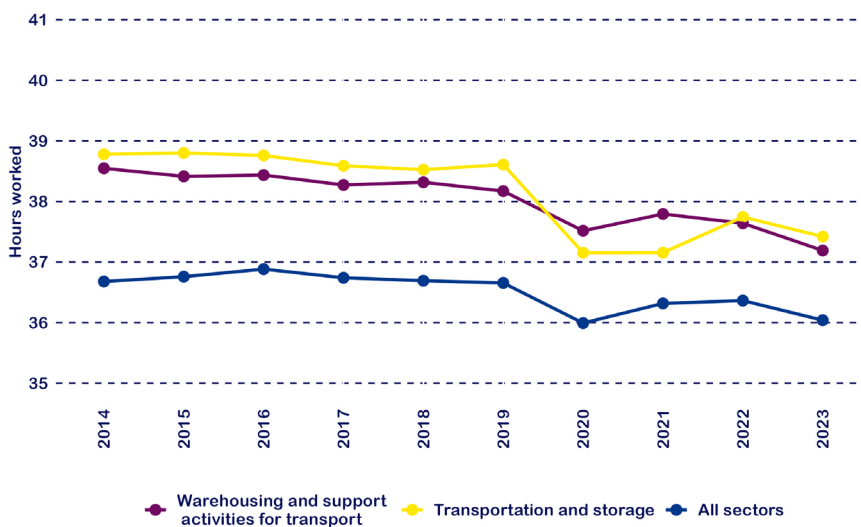
Figure 4 - Non-standard work schedules in the warehousing and support activities for transportation sector, the transportation and storage sector, and the overall economy in the EU-27, 2023



Source: Eurostat (2024). Labour Force Survey special extractions.

Employees in transportation generally work longer hours each week than employees in all sectors on average (Figure 5). Hours worked in the warehousing and support activities for transport sector are very similar to the transport sector on average. From 2014 - 2023, hours worked in the sector have slightly decreased from close to 39 hours, to just above 37 hours per week. This is around one hour per week longer than all sectors on average in 2023.

Figure 5 - Hours worked per week in in the warehousing and support activities for transportation sector, the transportation and storage sector, and the overall economy in the EU-27, 2014 - 2023



Source: Eurostat (2024). Labour Force Survey special extractions.

3.0 Influence of sectoral trends on labour market imbalances

Key determinants of labour market imbalances in relation to sectoral trends:

- Privatisation has led to reduced wages and benefits across sectors (aviation, water transport, railway), making jobs less attractive and contributing to labour shortages.
- Cost-cutting measures under privatisation, including reduced investments in training and development, widen the skills gap and create shortages in both technical and non-technical roles, especially in warehousing and aviation.
- Privatisation leads to the hiring of more cost-effective, but less experienced, workers in temporary roles, leading to higher turnover in the warehouse sector.
- The growth of e-commerce has shifted labour demand towards distribution centres and warehouses, driving demand for labour in areas like supply chain management and last-mile delivery. However, these rapid developments have raised concerns over productivity pressures and precarious employment conditions.
- The liberalisation of the airline and airport markets has led to cost-cutting measures, particularly in ground handling services, where labour costs represent around 80% of operational expenses.
- The transition to private port operators has led to staff reductions, with more reliance on temporary or contract-based workers.

Privatisation and competition

Privatisation and competition processes have impacted the sector of support activities for transportation. In water transportation, the corporatisation and privatisation of European ports have affected onshore labour (Notteboom and Vitellaro, 2019). The contemporary port environment is shaped by new dynamics, with operations shifting from processes that require significant capital investment to those that rely more on labour (Vaggelas and Leotta, 2019). The abolition of (full) labour pool systems in many Member States¹ inefficient, has granted private terminal operators full control over recruitment, allowing them to hire staff depending on their needs (Turnbull and Wass 2007; Van der Lugt et al. 2013 as cited in Notteboom and Vitellaro, 2019). In some ports, pool systems are still in place. With increasing trade volumes, employers have prioritised efficiency by investing in equipment and changing labour organisation and human resources practices. These developments compound existing challenges in health, safety, and security, as ports are regarded as one of the most dangerous work environments (Turnbull and Wass, 2007 as cited in Notteboom and Vitellaro, 2019).

The drive for cost efficiency has also resulted in the prevalence of non-standard and atypical employment, such as part-time and temporary roles for less skilled and experienced workers (Satta et al., 2019). Moreover, the overall number of dockworkers has decreased, increasing the physical and mental stress for those remaining in the workforce (Notteboom and Vitellaro, 2019). Additional challenges include uneven market liberalisation across Member States, lingering public sector dominance, national monopolies, and inconsistent regulatory frameworks (European Commission: Directorate-General for Mobility and Transport, 2024).

In railway transport, the shift from public to private entities has impacted labour conditions in both mobile professions and in support services for transportation. Given the sector's complexity, liberalisation started later than in other industries and remains an ongoing process (Broughton et al., 2024).

¹ A labour pool system is a structured arrangement commonly used in sectors like ports, transportation, and logistics, where a centralised entity (often a public or quasi-public body) manages a collective workforce available to multiple employers within the sector. Workers in the labour pool are typically assigned tasks or shifts as needed by different employers, based on demand, seniority, or rotation.

In aviation, the liberalisation of the airline and airport markets has yielded similar effects on labour markets as in other sectors. Lowering labour costs remains a key priority in labour-intensive sectors to enhance competitiveness, often achieved by streamlining production processes or reducing employment costs (Aviation Services Association World, 2024). Ground handling service providers, in particular, face substantial cost pressures, as labour accounts for approximately 80% of their operational expenses (European Commission: Directorate-General for Mobility and Transport, 2024).

Stakeholders consulted through the focus group for this study highlighted significant challenges in attracting and retaining ground staff and frontline workers in the transport sector since liberalisation processes started. According to them, such positions were regarded as stable and desirable, offering good benefits and job security. However, the liberalisation of the market introduced harsh competition among companies. This led to widespread workforce mobility, with employees frequently changing employers and losing many of the benefits they had previously enjoyed. The resulting uncertainty has been a source of concern for workers and has contributed to the decline in job attractiveness.

Stakeholders also noted the impact of low-cost carriers on working conditions. Passengers' dissatisfaction with limited services, such as restrictive baggage policies and additional fees for preferred seating, often results in frustration directed at frontline staff. This creates a challenging environment for workers who require training in conflict management to handle these situations effectively. Furthermore, the harsh working conditions, coupled with a lack of long-term career incentives, have demotivated younger workers, making it difficult to retain talent.

Stakeholders noted that the air traffic management sub-sector has not gone through cross-border liberalisation. In fact, only a few countries have opened the provision of airport air traffic management services to competitive tendering (Spain, the UK, Germany, Sweden and Norway) (Arblaster & Zhang, 2020). Stakeholders pointed out the relationship to air traffic management unions being strong, and employment and conditions being much more protected than in other occupations. An identified barrier to liberalisation of air traffic control services is the access to qualified air traffic controllers (Arblaster & Zhang, 2020). Related to this skills shortage, stakeholders believe privatised providers will not provide the strategy and long-term planning needed from the provider side to sustain a safe and efficient network.

The warehousing and storage industry has not experienced the same effects of privatisation as the occupations in the support services for transportation. This is largely due to its closer ties to private sector logistics and retail operations, which have historically remained outside direct public ownership. As a result, employment patterns, working conditions, and union representation have followed a different trajectory, yet face similar challenges in terms of working conditions, employment structure and wages.

Digital transition

Digitalisation and automation have significantly transformed the warehousing labour market, offering both opportunities and challenges. While automation offers cost reductions for employers, it risks job displacement particularly for lower skilled workers (European Commission: Directorate-General for Mobility and Transport, 2024; Team Hub, 2024). Furthermore, technologies such as robotics and automation enhance efficiency and safety by reducing injuries associated with heavy lifting, minimising falls, and alleviating worker fatigue (European Agency for Safety and Health at Work, 2024). However, digitalisation does not fully address health and safety concerns and workers are increasingly experiencing stress and fatigue due to intensified surveillance, algorithmic management, and repetitive tasks designed to maximise productivity (Team Hub, 2023; ETF, 2024c). Digital tools are used to control various aspects of workers' performance, including work process, breaks, conversations, bathroom visits and the number of goods stowed, picked and packed (Team Hub, 2024). Warehouse workers are reporting privacy concerns, high levels of illness and physical injuries stemming from monitoring, productivity pressure and surveillance (ETF, 2024c).

Moreover, the rising reliance on technology has created a shortage of skilled 'knowledge workers', who have high levels of professional knowledge, education or experience, with their work centred on creating, transferring and applying this knowledge (Surawski, 2019). In warehousing, there is a lack of such workers proficient in data-driven logistics and supply chain tools, which compounds the existing labour shortages for such roles in this sector (Descartes, 2024). High staff turnover in the sector and limited worker representation further expose workers to exploitation highlighting the need for balanced approaches to digital integration in warehousing (ETF, 2024c).

The rail sector is facing challenges in hiring workers with necessary skills, particularly as these skills are increasingly related to the green and digital transitions (Broughton et al., 2024). Digitalisation in the rail sector is starting to affect more complex tasks ranging from the maintenance of rolling stock to infrastructure development. These digital tools are progressively substituting lower technical skilled employment such as wagon inspection, break tests, coupling, and traffic management (Broughton et al., 2024). Other staff groups will have to acquire additional technical and IT skills. While digitalisation has the potential to enhance capacity, efficiency, and safety across all railway services, certain lower skilled activities still pose risks, even with the introduction of digital technologies. Due to new innovation in the automobile industry and new aviation technologies like electric vertical take-off and landing, airplane mechanics are increasingly sought after (Wildes, 2022).

In water transport sector, digitalisation is shifting labour in the port industry from low and medium-skilled roles to higher-skilled positions. This transition presents challenges for current port personnel to remain relevant and creates higher demand for skilled workers, with difficulties in meeting that demand (Broughton et al., 2024 and Vaggelas and Leotta, 2019). This new technological paradigm enhances various port operations, improving efficiency, safety, and energy savings in the maritime-port sector. Digitalisation drives these improvements, while automation ensures standardised operations, resulting in higher productivity, and improved quality across port processes (Vaggelas and Leotta, 2019). Workers are now able to work from lower-risk environment such as secure compartments and control rooms rather than directly on the quays (Broughton et al., 2024). By 2040, nearly 85% of tasks are expected to be automated, up from 27% in 2017 (Notteboom and Vitellaro, 2019). Therefore, the nature of activities is shifting from monotonous manual labour toward supervisory roles using sophisticated technology.

Trade unions emphasise the need for a balanced approach to automation and digitalisation to avoid discouraging new potential workers and causing job insecurity among current employees (ETF, 2023c). For many dockworkers, automation is seen as a threat as it is focused on cost savings rather than job creation (WEF, 2024). However, there can be many benefits if ports were to use cost savings to reinvest in job creation and climate-resilient projects resulting in long-term benefits to workers and communities (WEF, 2024).

The digital transition may help explaining the surplus of freight handlers (Table 1) reported by fourteen EURES countries. This trend is not due to less overall employment in the sector, as employment has increased until 2024 and remained steady throughout 2024 (Eurostat, 2024b). Instead, it is potentially related to the trend of digitalisation moving labour demand from lower to higher skilled jobs. Indeed, digitalisation in warehousing and support activities for transportation can reduce the reliance on manual labour in roles such as freight handling that are being automated by integrating technology. Examples include autonomous vehicles, drones, AI-powered sorting and optimised container handling processes in ports and warehouses (Paddeu et al., 2019). In response to digitalisation, manual workers such as freight workers will need to upskill IT and transversal skills (Broughton et al., 2024).

Rise of e-commerce

The demand for jobs in the warehousing and storage sector is projected to continue increase due to the continued growth of e-commerce, which drives the need for order fulfilment and returns processing (European Commission, 2023a). The rise of e-commerce has shifted the weight of work from typical retail infrastructure such as stores, towards distribution centres and warehouses (European Commission, 2023a). The proportion of e-shoppers rose from 53% in 2010 to 75% in 2023, marking a 22-percentage points increase (European Council, 2023). The e-commerce turnover increased from EUR 637 billion to a projected EUR 958 billion in 2024 (E-Commerce Europe, 2024). Additionally, new job opportunities are anticipated to arise in areas such as supply chain management, inventory management systems, and last-mile delivery (European Council, 2023).

Although, the growth of the e-commerce market in Europe suggests potential for increased sector-specific employment, tracking employment trends in e-commerce is challenging due to lack of data (Team Hub, 2023). E-commerce companies often adopt a multi-channel approach, integrating e-commerce with other channels, which complicates the collection of specific, disaggregated, and comparable data. Additionally, the high rate of temporary and precarious work contributes to significant fluctuations. Reporting on e-logistics employment trends is also difficult, although Germany, Italy, and France have reported overall job increases in this sector (Team Hub, 2023).

4.0 Recent global crises: implications for the labour market

Key determinants of labour market imbalances in relation to recent global crises:

- The COVID-19 pandemic led to massive layoffs in transportation support services during low demand periods.
- During the pandemic, many workers left the transport support services industry for alternative sectors, making it difficult to quickly rehire.
- The pause in recruitment and training during the pandemic caused a long-term shortage of skilled workers.
- Aviation and maritime sectors faced recruitment challenges as transportation levels returned to pre-pandemic volumes.
- The warehousing sector increased employment during COVID-19.

During the pandemic, support services for transportation experienced massive layoffs due to low levels of transportation. With the return of pre-pandemic levels of transportation, there has been severe staff shortages as workers had found work in alternative industries.

Although problems in the aviation system have existed long before COVID-19, the pandemic played a significant role in both causing and exacerbating the airport ground workforce shortage (Wandelt and Wang, 2024). Following the significant workforce reductions during the COVID-19 pandemic, the industry's primary concern is how to effectively attract sufficient ground workforce employees back to the aviation sector (Wandelt and Wang, 2024). In the summer of 2022, the situation in airports received significant media attention as labour shortages impacted passenger travel and ground crews went on strike (Politico, 2022; ITF, 2022). Since the recovery from the COVID-19 pandemic, airports have found it particularly hard to recruit baggage handling and security personnel. Trade unions called for more established work schedules, less outsourcing of airport operations to companies like ground handling service providers, and better wages (EU-OSHA, 2024). Despite some support measures and mechanisms, the pandemic put ground handling workers under great pressure to accept lower pay and worker conditions (Eurofound, 2022). To recruit a substantial number of new employees, the industry must tackle job uncertainty and dispel the post-pandemic perception of aviation as an unstable field. This effort will also have to focus on contract types, durations, and compensation (Wandelt and Wang, 2024).

Ground handlers estimate the permanent loss of staff across Europe to be around 32 000 - 40 000 people (25%) (Eurofound, 2022). On the other hand, air traffic controllers (ATCs) did not experience much workforce reduction as a consequence of the COVID-19 pandemic. This can be attributed to their essential role, which is protected by legislation that mandates their presence in airspace and airports regardless of traffic levels (Eurofound, 2022). Additionally, ATCs are challenging to replace due to the extensive two-year training period required to qualify for the role (Eurofound, 2022).

The rise of e-commerce which started during COVID-19 has also increased workload pressure consistently across freight in both support services and the warehousing sectors (EU-OSHA, 2024). During the pandemic, warehousing staff experienced an increase in employment. After the pandemic, labour shortages for some warehousing profiles arose as employees returned to their pre-pandemic employment sectors. Some of these shortages have yet to be resolved. Although it was anticipated that these shortages would ease after the pandemic, they continue to affect service levels and operational performance in the sector (Descartes, 2024).

Stakeholders consulted through the focus group provided insights into workforce challenges that have arisen in the aviation industry due to the COVID-19 pandemic and the ongoing war in Ukraine. Participants highlighted how both crises caused prolonged uncertainty, which significantly disrupted operations and workforce planning. Stakeholders emphasised the critical importance of training in the aviation industry, given the long duration required for roles such as ATCs, who need a minimum of two years (and up to four years in France), and mechanics, who require five years to become fully independent. They noted that pausing recruitment and training during the COVID-19 pandemic has caused a significant lag now impacting the sector. For example, countries like the Netherlands and Germany, the inability to maintain training led to severe staffing shortages in ground handling and security as traffic surged post-pandemic. In contrast, Austrian stakeholders prioritised training and recruitment during the pandemic, which supported a smoother recovery in the country.

The conflict in Ukraine has also created shifts in aviation traffic, increasing demand in regions such as Croatia, Austria, and the Balkans, while reducing traffic in Scandinavia. Stakeholders also emphasised the critical role of social tools and dialogue in retaining the workforce. For example, in Italy, agreements supported by the national unemployment insurance fund (e.g. Cassa Integrazione Guadagni) helped mitigate workforce losses. Luxembourg adopted similar strategies, which avoided layoffs and mitigated the impact of the crisis.

In the water transport sector, the COVID-19 pandemic, combined with a backlog of cargo at sea, has exacerbated the labour shortage crisis. Prior to the pandemic, the sector faced challenges such as an ageing workforce and limited recruitment and these issues worsened, with labour shortages in port operations and ground transportation leading to delays in cargo handling and extended dwell times. This has overwhelmed warehousing capacities, disrupting supply chains beyond the ports. As Moros-Daza and Jubiz-Diaz (2024) highlight, the shortage of workers has pushed the logistics system to its limits, affecting industries reliant on timely transportation and further straining global trade.

5.0 Workforce demographic and related issues

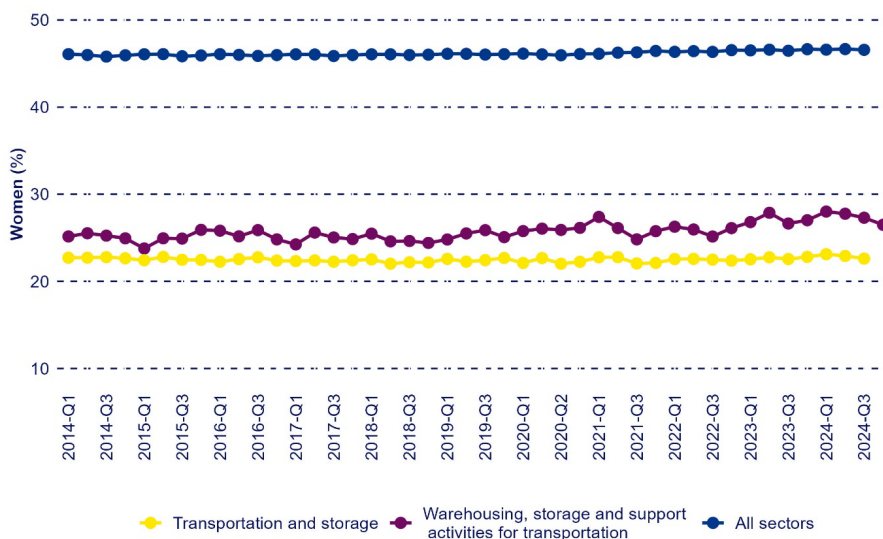
Key determinants of labour market imbalances in relation to workforce demographic:

- The transportation and storage sector faces challenges in replacing retiring workers, compounded by the health and capability needs of older employees.
- A lack of awareness about career opportunities and unattractive working conditions deter younger workers.
- Women are underrepresented, particularly in technical roles, due to gender stereotypes, limited career advancement, and inadequate workplace facilities.

The share of female employment in the warehousing and support activities for transportation sector stood at 26% in Q4 2024, around 20 percentage points lower than in the overall economy (47%). This gender imbalance is particularly pronounced in technical roles, where women remain severely underrepresented. Several factors contribute to this imbalance, including pervasive gender stereotyping, low-quality job opportunities, limited career advancement, and insufficient workplace facilities that support women's participation (EU-OSHA, 2024).

As seen in Figure 6, it is however higher than the overall transportation and storage sector on average (23%). The share of women employment in the warehousing and support activities for transportation sector has not significantly increased in the last 10 years (25% in 2014).

Figure 6 - Share of female employment in the warehousing and support activities for transportation sector, the transportation and storage sector and the overall economy in the EU-27, 2014 Q1 – 2024 Q4



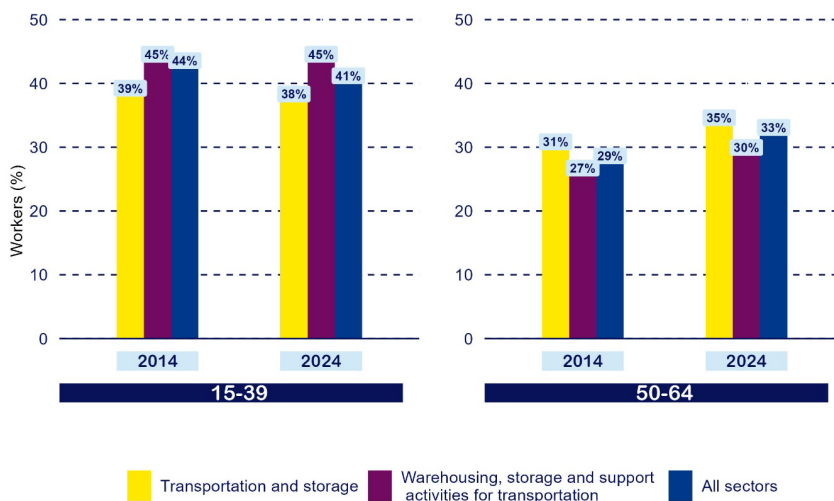
Source: Eurostat [lfsq_egan2, lfsq_egan22d]

The transportation and storage sector across the EU faces significant challenges due to an ageing workforce. This demographic shift creates difficulties.

Jobs in the warehousing and support activities for transportation sectors attract relatively more younger workers than the overall economy or the transportation and storage sector. In 2024, workers aged 15-39 years accounted for 45% of warehousing and storage workers, four percentage points more than the overall economy and seven percentage points more than the average in the transport sector. This share has also remained stable during the last 10 years (45% in 2014). However, younger workers are still often underrepresented in some areas of the sector largely due to a lack of awareness about career opportunities, unattractive working conditions, and a disconnect between employer expectations and the aspirations of younger generations (EU-OSHA, 2024; Proequity, 2024).

On the other hand, the share of workers aged 50-64 years in the warehousing and storage sector has increased by three percentage points between 2014 (27%) and 2024 (30%) (Figure 7). In 2024, they accounted for respectively five percentage points less than in the transportation and storage sector, and three percentage points less than the rest of the economy. The ageing workforce is a challenge as replacing retiring workers and addressing the health and capability needs of older employees (EU-OSHA, 2024).

Figure 7 - Employment age structure in the warehousing and support activities for transportation sector, the transportation and storage sector and the overall economy in the EU-27 in 2014 and 2024

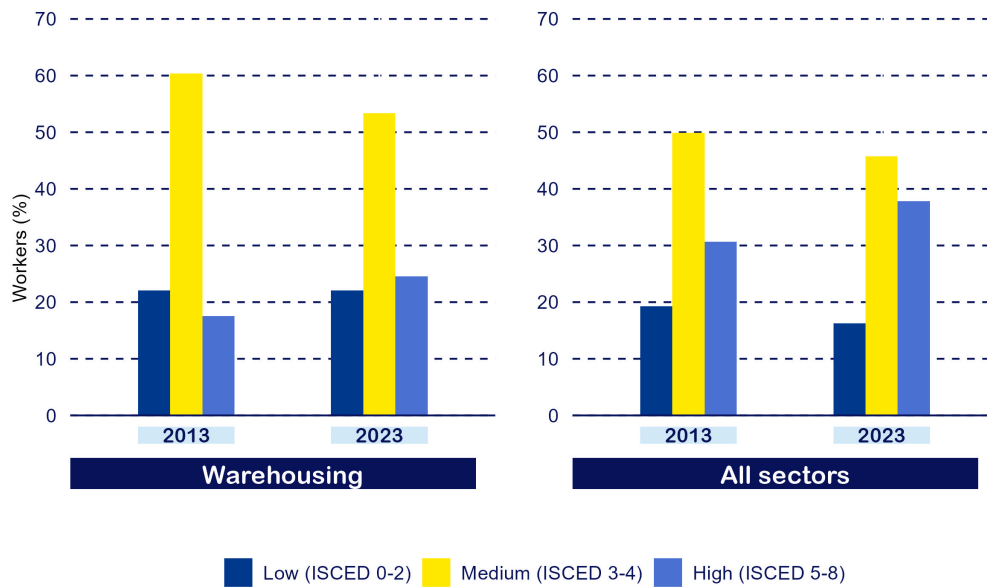


Source: Eurostat [lfsq_egan2, lfsq_egan22d]

The qualification profile of workers in the warehousing sector reveals a higher reliance on medium- and low-skilled workers compared to the overall economy. In 2023, 53% of the warehousing sector's workforce was medium-skilled, and 22% was low-skilled, compared to 45% and 16%, respectively, across all sectors. Hence, the sector employed relatively fewer high-skilled workers, with only 17% in warehousing versus 24% in the broader economy.

As seen in Figure 8, between 2013 and 2023, the proportion of low-skilled workers in the warehousing sector remained stable at around 22%. However, the share of medium-skilled workers decreased by seven percentage points, from 60% in 2013 to 53% in 2023, while the proportion of high-skilled workers increased by seven percentage points, from 17% to 24%.

Figure 8 - Education structure in the warehousing, storage and transportation activities for transportation sector and the overall economy in the EU-27 in 2013 and 2023



Source: Eurostat (2024). Labour Force Survey special extractions.

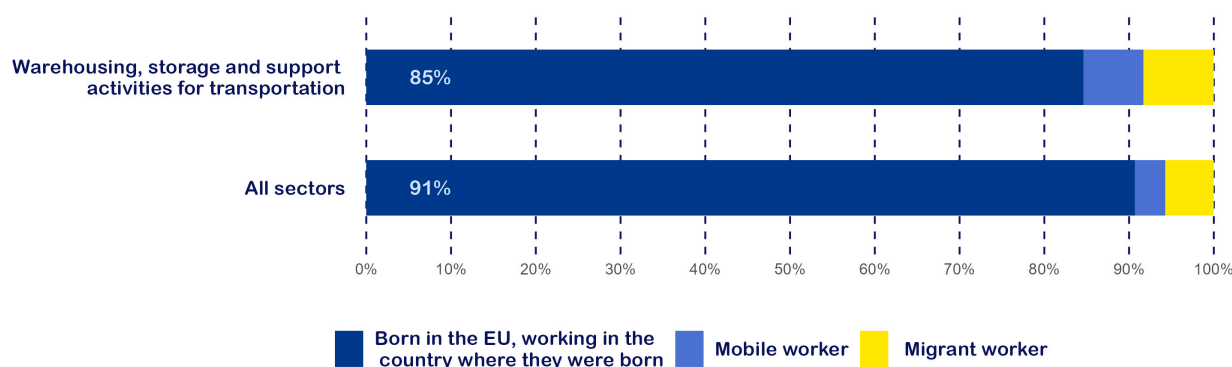
6.0 Labour migration and mobility

Key determinants of labour market imbalances in relation to labour migration and mobility:

- The warehousing and support activities for transportation sector relies less on EU nationals working in their home countries compared to other sectors of the economy, with higher reliance on mobile and migrant workers.
- Countries like Poland, Romania, and Bulgaria provide significant labour for warehousing jobs in destinations like the Netherlands, Czechia and Germany.
- Warehousing is one of the sectors with the highest number of posted workers within the EU, with increasing trends of third-country nationals being posted to work in various EU countries.

In 2023, the sector relied on EU nationals working in the country where they were born to a lower extent compared to all sectors of economy. Mobile workers represented around 5% of the overall employment while migrant workers around 9% (Figure 9).

Figure 9 - Overall employed by country of origin in the warehousing, storage, and support activities for transportation sector and overall economy, EU-27 2023



Note: Mobile workers refer to people originating from the EU, but working in another Member State. Migrant workers refer to people originating from outside the EU working in a Member State.

Source: Eurostat (2024). Labour Force Survey special extractions.

As reported by GiGroup (2022), a large staffing and HR company present in most European countries, logistics is one of the priority industries for international mobility. Occupations in the warehousing and storage sector such as warehouse operators, forklift drivers and specialised technicians are some of the most required job profiles. The report identifies the main sourcing countries of mobility flows as Ukraine (pre-Russian aggression), Poland, Bulgaria, Hungary and Romania. The key destination countries are Czechia, Germany, the Netherlands, Poland and Slovakia. In Poland, a survey of foreign employees shows that 20% of non-Polish nationals with Polish work permits work in logistics (Norman, 2024). Around 225 000 people in the Polish warehousing sector are not Polish nationals (Norman, 2024). The largest represented nationalities are Ukraine, Belarus, Georgia.

Consulted stakeholders from the Netherlands pointed out that the warehousing and storage sector in many European countries relies heavily on mobile workers, particularly from Eastern Europe. This is often because local labour forces are insufficient or unwilling to take up these roles. Countries like Poland, Romania, Bulgaria, and Spain have available labour that is recruited to fill positions in countries like the Netherlands, where the demand for warehouse workers can be high. Stakeholders in the aviation maintenance sector report shortages during peak season, attributing the lack of personnel to better wages in the Middle East.

To fill shortages in warehousing, some Member States struggling to find workers from inside Europe, like Belgium, are facilitating opportunities to move and work in Belgium for some sectors like warehousing (Januzi, 2023). A 2024 survey in Poland shows that housing offered by employers or employment agencies is very important to attract intra-EU movers or third-country nationals. Employers have noticed offering housing increases competitiveness and employee loyalty (Norman, 2024).

In addition to workers moving within the EU and from third countries, warehousing and support services for transportation is one of the sectors with the most posted workers (European Labour Authority, 2024). A posted worker is an employee temporarily assigned by their employer to provide services in another EU Member State. This practice is rooted in the EU's principle of free movement of services (European Labour Authority, 2024). The posting of third-country nationals within the EU labour market is an increasing trend in labour mobility. These workers often face heightened vulnerability due to their reliance on employers for work and residence permit renewals. As a result, they are at greater risk of labour rights violations, exploitative working conditions, and irregular social security contributions (European Labour Authority, 2024).

7.0 Working conditions and impact on industry's reputation

Key determinants of labour market imbalances in relation to working conditions:

- Harsh working conditions, including low wages, unsociable hours, and physically demanding tasks, contribute to high turnover rates in the warehousing and support activities for transport sectors.
- Poor job stability, exacerbated by temporary contracts and subcontracting, makes it difficult to retain workers.
- The rise of automation and digital tools in sectors like maritime and aviation raises concerns about job security, deterring new recruits and contributing to workforce shortages.
- Inconsistent working conditions and lack of career progression discourage younger workers from staying in sectors.

Deteriorating working conditions

Jobs in warehousing and transportation support services face significant image challenges, with working conditions being a key contributor to labour shortages affecting both low- and high-skilled workers (European Commission, 2023b). Worker retention is particularly problematic as younger employees are often quick to leave due to the prevalence of low-skilled roles, shift work, and harsh working conditions in some sectors (Broughton et al., 2024). The negative public perception is also shaped by unsocial working hours, insecure contracts, and low wages (Broughton et al., 2024). This sector is also viewed as highly demanding and physically challenging (Proequity, 2024). The port sector, for example, is frequently perceived as dirty, heavy, hard to access and unsociable, making it difficult to attract talent (SEOR, 2024).

In airports, the irregular hours and pressure of the ATC profession also discourage applicants (Le Forem, 2024). Physically demanding roles in the aviation sector, like ground handling, have become less appealing as wages decrease and working conditions worsen (European Commission: Directorate-General for Mobility and Transport, 2024). In addition to long hours and night shifts, ground handling workers at airports often face split shifts, which require them to work shorter periods spread across multiple days, leading to more time lost in commuting (ETF, 2023b). Alongside retention, the attractiveness of these jobs is considered one of the sector's major challenges (Broughton et al., 2024).

Stakeholders consulted through the focus group have expressed concerns about the negative public perception of ATCs. Airlines have emphasised their role in contributing to flight disruptions, fuelling this perception. As a result, the ATC profession is increasingly viewed as controversial and unstable, complicating recruitment efforts. Furthermore, there are significant concerns about the potential for AI to replace ATCs in the next 20 years. This has created uncertainty for potential recruits who are reluctant to pursue a career in a field they believe may not have long-term viability, thus exacerbating recruitment challenges.

Consulted stakeholders in the warehousing sector have highlighted how the rise of digital platforms is contributing to deteriorated working conditions through the growing trend of subcontracting and the fragmentation of supply chains. According to them, while digitalisation can help address environmental issues, it has led to the deterioration of working conditions and wages, particularly at the lower ends of supply chains where monitoring is more difficult. This has resulted in unattractive working conditions that contribute to labour shortages. In the e-commerce logistics sector, the complexity of longer supply chains has made it more challenging to ensure ethical standards are met, exacerbating issues related to working conditions.

In occupations in warehouses, some companies have raised salaries to address labour shortages and fill positions. However, this approach is not widespread across all warehouse and transportation support activities. Strikes to increase wage conditions have taken place in airports both in Europe and internationally (European Agency for Safety and Health at Work, 2024). It should be noted that warehousing and support services for transportation have the highest wage-adjusted labour productivity ratio out of all the transportation sector. This means employees in this sector add high amounts of value compared to how much they cost (Eurostat, 2024a).

In ground handling, the average contract length has noticeably decreased, and frequent changes in service providers exacerbate the issue. When operators switch providers, the new employer often imposes revised terms that further degrade working conditions (ACI Europe et al., 2018). Low salary levels make it relatively easy for competing companies, including other ground handling firms, to attract employees with modest pay increases. Recruitment and retention challenges frequently result in a less experienced workforce with lower skill levels, or workers requiring more extensive training, which can ultimately impact performance (Aviation Services Association World, 2024). Aviation mechanic jobs are not attracting new entrants to the field because this occupation is considered physically demanding and not attractive compared to other jobs. Shortages in the occupation means heightened pressure on current mechanics.

In the rail sector, deteriorating working conditions stem from both direct wage pressures and indirect measures, such as the removal of benefits, social insurance, or bonuses (ETF, 2023d). In Germany, cases have emerged where workers were transferred to subsidiaries offering lower wages and fewer benefits (ETF, 2023d). Trade unions in Germany, Italy, and Austria have reported declining wages and income levels when companies opted out of sectoral collective wage agreements or adopted alternative agreements. These alternative agreements often align with the warehousing and logistics sector where wages and financial conditions are notably less favourable (ETF, 2023b).

Job security

Despite employment growth in the warehousing and support services for transportation sector, structural fluctuations tied to seasonal peaks highlight pervasive job insecurity across the supply chain (Team Hub, 2024). This insecurity is associated with several risks that negatively affect both workers and operations, including insufficient skills and training, inadequate familiarity with the work environment, limited access to personal protective equipment, and poor health and safety monitoring (Team Hub, 2024).

The growing emphasis on automation and digitalisation in the maritime sector, including remotely operated navigation and fully automated vessels, presents a challenge to its appeal as a career choice. While these advancements aim to enhance efficiency, concerns among current workers about job security may deter potential newcomers who view the sector as increasingly reliant on technology and less on human labour (ETF, 2023a).

Health and safety

When it comes to occupational health and safety, the transportation and storage sector is among the most hazardous industries in the EU, accounting for 15% of all fatal workplace accidents in 2020 and reporting a significantly high rate of non-fatal incidents – 2 351 cases per 100 000 workers compared to 1 444 in the overall economy (EU-OSHA, 2024). Workers in this sector face substantial physical risks, including prolonged sitting, repetitive tasks, vehicle accidents, heavy lifting, and exposure to hazardous chemicals and extreme weather conditions. They also encounter psychosocial risks such as time pressure and isolation (EU-OSHA, 2024). Notably, 61% of organisations in the transportation and storage sector have health and safety representatives, particularly in larger companies, which is well above the EU-27 average and reflects the elevated risks in this sector (EU-OSHA, 2024). However, growing labour shortages are increasing the strain on existing staff, worsening health, safety, and psychosocial risks (ETF, 2023b).

In the aviation sector, ground staff face numerous health and safety challenges, including exposure to noise, vibrations, ultrafine particles, long working hours, physically demanding tasks, and night shifts (ETF, 2024a; Eurofound, 2022; EU-OSHA, 2024; European Commission: Directorate-General for Mobility and Transport, 2024). A recent investigation in the Netherlands, representative of risks in the profession, highlighted severe health hazards faced by baggage handlers at Schiphol Airport (Amsterdam). Excessive physical strain from heavy lifting has led many handlers to develop musculoskeletal injuries, such as chronic back pain, torn muscles, and, in some cases, permanent disabilities. Company doctors have warned that up to half of the workforce could develop occupational diseases (NL Times, 2023).

Aviation cargo and workers in warehouses face similar occupational health risks, including musculoskeletal disorders, such as muscle strains, chronic lower back pain, and shoulder injuries, exacerbated by confined workspaces, repetitive tasks, and high-pressure environments (Aviation Services Association World, 2024; Team Hub, 2023). In the warehousing sector in particular, workers also face significant psychosocial risks, including interactions with difficult customers, time pressures, and irregular, long, or night working hours (Team Hub, 2023).

Workers are also exposed to hazardous gases, such as pesticides from fumigated freight containers which can lead to respiratory issues and long-term health problems. Poor ventilation and extreme temperatures - both heat and cold - further contribute to health risks such as heatstroke, cold stress, and the exacerbation of musculoskeletal disorders (European Agency for Safety and Health at Work, 2024).

Health and safety are also affected by the growing use of digital and automation tools for monitoring productivity. These tools, including social rating systems and non-transparent algorithms, can increase workloads and stress levels (Broughton et al., 2024). Surveillance technologies, such as cameras and wearable devices tracking warehouse workers' movements, are becoming more invasive (ETF, 2024d). A UK survey by worker rights platform Organise (2018) found that 74% of surveyed Amazon workers refrained from using the toilet to avoid performance-related warnings, while 55% reported experiencing depression since starting their employment (Eurofound, 2020). While GDPR offers some protections, trade union representatives have emphasised the need for stronger legislation and better enforcement to safeguard workers' privacy and rights. Trade unions have also warned that AI and algorithms in logistics often prioritise profitability over workers' conditions, advocating for enhanced legal frameworks to ensure these technologies benefit workers rather than exploit them (ETF, 2024d).

The role of social dialogue

The warehousing and storage occupations across Europe faces significant challenges related to collective bargaining, worker representation, and the regulation of working conditions. As highlighted by Broughton et al. (2024), the negotiation of collective bargaining agreements (CBAs) in the road transport and logistics sectors is particularly complex. In many EU countries, trade unions struggle to represent workers, especially migrant workers, and those from disadvantaged backgrounds. These workers often lack awareness of their rights, which exacerbates the difficulties of improving working conditions. The absence of coherent national or EU-level policies regulating new urban mobility services and logistics further complicates efforts to create fair and motivating work environments (Broughton et al., 2024).

On the support services for transportation side, the air traffic management (ATM) sector offers an interesting contrast, as workers, particularly air traffic controllers (ATCOs), enjoy high salaries and strong bargaining power due to the specialised training required for their roles. This, along with the crucial safety tasks they perform, makes ATCOs more unionised and more effectively represented by trade unions (Eurofound, 2022). However, the ground handling sector in civil aviation faces a more fragmented union presence due to factors like the diversity of employers, irregular work schedules, and the prevalence of migrant workers. These barriers make it harder for unions to organise and protect the rights of ground staff (Eurofound, 2022).

Related to warehousing occupations, the landscape is similarly fragmented and presents considerable challenges to collective bargaining efforts. As noted by Team Hub (2023), unionisation in the e-commerce and logistics sectors is often weak, with obstacles including a lack of company-level representation, high workforce turnover, and the prevalence of precarious work. Moreover, the presence of migrant workers complicates unionisation efforts due to linguistic and cultural barriers. The role of migrant workers in the sector is crucial, as they often fill low-paid, precarious jobs. However, their vulnerability makes them an even more challenging group for unions to organise and advocate for. This situation is exacerbated by the lack of national coordination and support for union activities, which remains a key objective yet to be achieved in most countries (Team Hub, 2023).

Countries like Italy and Spain have seen some success in collective bargaining efforts, but this remains the exception rather than the rule. Overall, fragmentation is the norm and union efforts to negotiate collective agreements have largely failed for example in Germany, France and Poland (Team Hub, 2023). Some positive examples include grassroots campaigns in Germany and Poland and strategic litigation in demonstrating bogus self-employment (Spain), unlawful obstruction of a labour dispute (Poland) and enforcing compensation for sick leave, accidents at works and COVID-related measures (France) (Team Hub, 2023). To improve conditions for collective bargaining, unions and companies need to build social dialogue, trust, and mutual recognition, with support from policymakers.

8.0 Skills and qualifications gaps

Key determinants of labour market imbalances in relation to skills and qualifications gaps:

- The evolving job requirements, driven by digitalisation and increasing complexity, demand advanced technical skills and higher education levels. Currently, there are skills mismatches.
- Insufficient training infrastructure, slow adaptation of educational programmes, and a lack of in-house training opportunities hinder the development of essential skills and exacerbate skills mismatches.
- The rail, aviation and maritime sectors face significant skills gaps in the transition to green technologies, requiring specialised training to meet climate goals and address the increasing demand for STEM and communication expertise.

Many jobs in warehousing and support activities for transportation are undergoing significant evolution, requiring workers to transition from lower or moderately skilled roles to those demanding more advanced technical skills (SEOR, 2024; Broughton et al., 2024; Vaggelas and Leotta, 2019; EU-OSHA, 2024). This shift in skill requirements is a key factor contributing to labour shortages across these sectors. Addressing the skills mismatch necessitates better alignment between educational programmes and industry needs to equip the workforce for emerging demands (TLF, 2024).

In the port sector, digitalisation is increasing the complexity of work and raising the demand for higher education levels (SEOR, 2024). The number of technical onshore jobs is rising and is expected to continue growing, potentially creating opportunities for underrepresented groups in the workforce (Pike and Terpilowski, 2022). To support this shift, investments in training for the twin transition are essential. The European Commission highlights that insufficient infrastructure and training facilities in some countries act as bottlenecks, limiting access to maritime academies (ETF, 2023b).

Addressing both current and future skills shortages requires integrating specialised talent into the sector. The European Economic and Social Committee (2024) reported that many businesses struggle to provide in-house training and professional development opportunities even if it is crucial to adapt their employees' skills to advancements in green technologies and digitalisation. While ports recognise the importance of training, port entrepreneurs are hesitant to invest in training due to concerns about losing personnel to competitors (SEOR, 2024). Terminal operators face the challenge of updating training programmes to improve workers' adaptability and efficiency in managing the fluctuating demand and seasonal variations typical of port operations (Broughton et al., 2024).

In warehousing occupations, skills mismatches are evident, particularly in the lack of both technical expertise (e.g. technologies and regulations) and non-technical skills (such as language proficiency) or relevant qualifications for managerial roles (Condon et al., 2024; Le Forem, 2024). In the logistics sector, future skills requirements over the next five years emphasise the growing demand for specific expertise. The most sought-after skills include quality and safety at work, environmental knowledge, materials handling, and proficiency in IT and information systems (OPCO Mobilites, 2022).

Skills mismatches in the aviation sector often stem from language-related challenges, particularly for ATCs and flight dispatchers, as both roles demand a high level of language proficiency (Le Forem, 2024). ATCs also face high attrition rates during their minimum two-year training period (European Commission: Directorate-General for Mobility and Transport, 2024). Eurocontrol (2024) reports ATCs are understaffed due to difficulties in recruiting future talents (Directorate-General for Mobility and Transport, 2024, and ATCEUC and ETF, 2024). Representatives of the sector have raised concerns that without increased investment in training, the current pace will be insufficient to address the already limited capacity in this critical field (ATCEUC and ETF, 2024). Similar concerns apply to engineers and technicians, whose roles are expected to remain based on humans despite ongoing digitalisation and automation (ATCEUC and ETF, 2024). In terms of aircraft engine mechanics and repairers, stakeholders in the focus groups identified a lack of investment in education by the aviation industry into young mechanics. A mechanic needs five years of training to work on their own, meaning the shortage will not be resolved quickly. Mechanics currently spend much time in continuous training regarding digitalisation and greening, reducing the time they can spend on the ground. Due to older and more experienced mechanics leaving or retiring from aviation during COVID-19, there is a lack of mechanics and current ones are less skilled.

Stakeholders consulted through the focus group pointed out that while digitalisation has streamlined processes such as check-in and baggage handling, it has also created new roles, such as floor walkers and facilitators, to assist the less familiar customers with digital technology. These roles typically require lower skill levels and have led to an increase in lower-paying, less stable employment, often filled by temporary workers, including students. Stakeholders were concerned that without proper training and long-term workforce planning, the ongoing technological advancements could exacerbate existing labour shortages and make jobs even less attractive.

Despite being the most sustainable mode of passenger transport and its anticipated growth in the coming years, the rail sector faces significant challenges in addressing skills and qualification gaps. The transition to a climate-neutral economy demands specialised skills in sustainability and green technologies (Cuzzupe, 2023; Eurofound, 2023). Rail operations and infrastructure services are undergoing a transformation, with around 60 rail-specific profiles requiring updates in fundamental and transversal skills across all qualification levels (Staffer, 2024). Digital transformation further necessitates upskilling in areas such as STEM and communication (NewRail et al., 2019).

The Shift2Rail Joint Undertaking identifies two major challenges for rail companies: managing complex, data-heavy systems and building a diverse and motivated workforce. To address these, accelerating knowledge transfer is crucial, while ensuring a balance between academic-level education and vocational training (Shift2Rail Joint Undertaking et al., 2019). There is also an increasing need for continuous monitoring of skills intelligence which includes updating occupational profiles, curricula, and vocational programmes to align with evolving industry demands (Staffer, 2024).

According to a report written for the European Rail Skills Alliance, the industry's future lies in digitalisation, sustainability, and technology. Key competencies for modern railway professionals include systems engineering, cloud-based applications, data analysis, cybersecurity, and virtual reality simulators, alongside traditional expertise in mechanical and electrical engineering (Staffer, 2024). However, the current educational landscape is not sufficiently aligned with these needs. Across Europe, tertiary education programmes lack tailored pathways into rail sector positions (Cuzzupe, 2023).

To ensure employers and employees stay current with technological advancements, a standardised training system is needed to include everyone in the transition process. Automation, particularly Automatic Train Operation (ATO), is reshaping roles such as drivers and control staff. While ATO enhances efficiency and capacity, it also demands new skills and organisational adjustments (CER and ETF, 2022). Workforce planning and training are therefore critical to adapting to these technological shifts and to ensuring that automation complements a skilled and capable workforce.

In the maritime transport sector, the shift towards sustainability and reducing greenhouse gas (GHG) emissions is expected to require new measures and policies as expertise in technical areas such as electric ship batteries, onboard fuel cell power production, and innovative ship design becomes increasingly essential. While the production and chartering of low-emission vessels is underway, the adoption of renewable energy and low-carbon fuels remains limited, with 98.7% of fuel in use still derived from conventional oil and petroleum products (Directorate-General for Mobility and Transport, 2024). As the shipping industry transitions to alternative fuels, it is crucial for both onboard and onshore personnel to develop skills related to the handling and bunkering of these new energy sources (Oksavik et al., 2020).

9.0 Recruitment practices to fill labour and skill gaps

Key determinants of labour market imbalances in relation to recruitment practices:

- High reliance on subcontracting and temporary work employment creates fragmented workforces with poor conditions and job insecurity.
- Flexible and temporary contracts, such as zero-hours arrangements, undermine job stability and lead to high turnover rates.
- Wage discrepancies across subcontracting firms and sectors make positions unattractive.
- Seasonal and fluctuating demand in sectors like ports and airports limits permanent hiring, increasing dependency on temporary agency workers.
- Lack of training and skill development for subcontracted and temporary workers exacerbates safety risks and reduces retention.

Subcontracting and temporary work agencies are common across warehousing and support services for transportation. Subcontracting occurs when large firms delegate operational work to smaller companies, such as hiring an external company to handle maintenance or cargo handling (ETF, 2022). In contrast, temporary work agencies supply workers to perform tasks directly within the primary company. These contracting arrangements enable the primary company to decentralise operations while maintaining control over management, coordination, and economic decisions.

These temporary arrangements create unstable working environments, resulting in high turnover rates. In ports, the need for support services is highly dependent on fluctuating and seasonal cargo volumes. To manage these variations in demand, port terminal operators usually maintain a core workforce and supplement it with temporary dockworkers during peak periods. These additional workers are often sourced through employment agencies and, to a lesser extent, official labour pools managed by the ports, in accordance with national law on labour port schemes². The significant variability in transport demand requires adjustments to working practices, promoting flexibility and operational efficiency (Notteboom and Vitellaro, 2019). Temporary contracts are common for both starter and experienced employees to cope with business peaks. Experienced staff often seek more stable opportunities elsewhere, adding to the shortage of skilled staff in their sectors (Team Hub, 2023).

In railway and aviation (particularly for ground handling activities), larger companies increasingly rely on part-time contracts, zero-hours contracts, temporary work through agencies, and outsourcing to recruit workers (ACI Europe et al., 2018; ETF, 2022; ETF, 2023d).

² Dock labour schemes take many different forms. They are at times based on a centrally managed pool of registered dock workers or not. The use of such a pool for hiring can be mandatory by law or not. Dock labour pool schemes lay out that: there is an 'in group' of officially registered (licensed) dockworkers; dockworkers are not permanently hired by terminals and are obligated to source work through the central pool; there are unemployment benefits for dockworkers in the pool during shortage of work (Notteboom, 2020).

The outsourcing trend is particularly concerning, with regards to jobs once seen as central in the railway industry, such as maintenance, now often handled by external contractors. These outsourced workers typically face worse conditions, receive lower pay, which compromises safety and the quality of their work (CER and ETF, 2022). According to trade unions, the outsourcing of tasks fragments the railway forces into a second peripheral labour market of railway services often with low wages, high job insecurity and bad working conditions (ETF, 2023d). In Sweden for example, light and heavy maintenance is mostly outsourced to either subsidiaries of larger rail companies or external partners. These act as staffing companies, with the associated working conditions, rather than railway undertakings (ETF, 2023d). France is another example where outsourcing is taking place in fields of material and track maintenance, station services and safety (ETF, 2023d).

Consulted stakeholders from the Netherlands emphasised that much of the work in the warehouses is part-time. Many roles are filled by students taking on side jobs or by individuals balancing other commitments, such as childcare or a second job. Often, part-time contracts are not employee preference, but rather what employers offer. This makes part-time and flexible work arrangements a defining characteristic of the sector. Stakeholders point out that flexibility often comes at a cost. In the Netherlands, around 60% of workers in warehousing and logistics operate under flexible contracts. Zero-hours contracts and on-call work are particularly prevalent, making income stability a concern for many workers.

The impact of subcontracting varies significantly across the EU. In certain cases, subcontracting is viewed as having adverse effects on social conditions and, in some instances, as facilitating social dumping. These challenges arise due to discrepancies in how labour rules are applied, which may not guarantee equivalent conditions, particularly in areas such as social security or pension schemes (ACI Europe et al., 2018). Furthermore, subcontracted workers often fall under different collective agreements based on the activity type, and in some airports, they are excluded from collective agreement altogether, a situation also affecting temporary agency workers. Additionally, trade union membership rates are typically lower among subcontracting firms, further contributing to the divergence in social conditions (ACI Europe et al., 2018). A related practice is bogus self-employment, where workers are technically self-employed allowing firms to avoid paying social benefits, yet employees do not fulfil the conditions of self-employment like setting their own working hours. Overall, subcontracting, bogus self-employment and temporary labour contracts suggest workers are in precarious situations in the warehousing sector (ETF, 2022; Team Hub, 2023).

In airports, the extent of subcontracting varies widely depending on the location. During peak periods, such as the summer season, the proportion of agency workers in ground handling can rise significantly, as seen in the Netherlands (Eurofound, 2022). In Spain, nearly half of the ground handling workforce is employed on a temporary basis throughout the year. This seasonal increase underscores the critical role of temporary agency workers in meeting the fluctuating demands of the ground handling sector (Eurofound, 2022). In Frankfurt, for example, the airport employs 880 subcontracted workers (out of around 80 000), primarily in cleaning services (750), as well as in transport (70) and pushback/towing services (60) (ACI Europe et al., 2018 and Fraport, 2024). In Berlin and Hamburg, subcontractors make up about 10% of the workforce while Vienna relies on subcontractors during peak periods. In certain countries, legal frameworks limit subcontracting, such as Italy's restriction to a maximum of 30% of activities (ACI Europe et al., 2018).

In several countries, including Italy and Belgium, warehouse and logistics companies exploit differences in wage-setting mechanisms to reduce labour costs. In Belgium, for instance, some businesses engage in 'shopping' between joint committees to find sectors with less protective conditions, undermining minimum wages and rights established through collective bargaining (Team Hub, 2023).³ Similarly, in countries with statutory minimum wages, social dumping remains a significant challenge, particularly for small subcontracting companies where enforcement of labour rights is weak. For self-employed subcontractors, minimum wage protections are often non-existent. Poland stands out as an exception, where all workers, including the self-employed, are entitled to a universal statutory minimum wage, though the amounts remain modest (Team Hub, 2023).

³ A joint committee is an advisory body comprising representatives from both the employers' union and the employees' union. Its primary purpose is to negotiate collective labour agreements which set out terms for working conditions, minimum wages, bonuses, and other employment-related matters within a specific sector ((Employment Belgium, 2024). In the warehousing and support services for transportation, the joint committees 140.03 (transport and logistics for blue-collars), 226 (transport and logistics and international trade for white-collars) apply.

10.0 Measures to tackle labour market imbalances

Key determinants of measures to tackle labour market imbalances:

- Training and skills development initiatives focus on upskilling, reskilling, and industry-specific programmes to align workers' skills with labour market needs.
- Strengthening collaboration and partnerships involves engaging with universities, vocational schools, policymakers, and businesses to address labour shortages.
- Improving workplace conditions and benefits helps attract and retain workers by enhancing safety, job security, wages, benefits, and flexible work arrangements.
- Enhancing industry image and attractiveness involves increasing sector visibility, fostering community engagement, improving transport access, and upgrading workplace amenities.
- Promoting gender equality and diversity includes implementing mentorship programmes, equal pay initiatives, flexible policies, and bias-free recruitment processes.
- Advancing technology and automation helps reduce workload, increase efficiency, and improve job quality through robotics and digital solutions.

To address labour shortages across various sectors, industries are implementing a range of strategic measures focused on improving skills, enhancing workforce mobility, promoting inclusivity, and creating more attractive working conditions.

Strengthening training initiatives

Implementing timely upskilling and reskilling initiatives is expected to enhance the retention of airport ground employees (Wandelt and Wang, 2024). In the ground handling sector, the European Union Aviation Safety Agency (EASA) has proposed regulations, expected to be published in early 2025, including mandatory higher levels of training amongst defined operational procedures and a robust safety culture (EASA, 2024). By prioritising training, regulation in the sector aims to equip workers with the skills necessary to meet operational demands and create a safer and more attractive work environment (Aviation Services Association World, 2024 and Uphues, 2024). However, stakeholders note that the three-year implementation period may motivate employers to start reforms later rather than earlier.

The port sector also identifies training as a necessity. Stakeholders are urging policymakers to provide future-proof training, education and lifelong learning opportunities with workers in the centre of the digital and green transition (ETF, 2024b). The ETF positions port workers as essential 'front line' workers vital to keep the world moving (ETF, 2024b).

Related to training, companies in the warehousing sector are using training policies, amongst other strategies to attract workers. Since there are also skills shortage gaps, companies are implementing internal training policies and collaborating with universities and technical schools to align company and skills standards (GiGroup, 2023). For example, all employees at DHL Global Forwarding carry out a ['DHL Certified International Forwarder'](#) or ['DHL Certified Freight Specialist'](#) programme to ensure relevant skills for their professional activity (DHL, 2021). [CEVA Logistics](#) collaborates with vocational schools and universities to facilitate the entry of young talent into the warehousing sector, addressing both labour shortages and qualifications (CEVA Logistics, 2024).

All industries can benefit from collaboration with schools and companies. For example, in the warehousing sector in Germany, there are multiple high school courses on the topic of the logistics sector, and cooperation with universities to promote industry visibility (GiGroup, 2023).

Improving sector image and attractiveness

To structurally solve labour shortages by increasing retention, work stability, safety and overall well-being of employees need to be improved (Wandelt and Wang, 2024). Improving work environments is a key strategy for employee retention, with a focus on comfort and safety (EVA, 2016). Investments are directed towards upgrading building features such as lighting, ventilation, and safety measures. New developments are often designed with these features in mind, while existing facilities are being retrofitted to enhance the workplace environment (Prologis, 2019). Overall, competitive wages and benefits are the main strategies available to companies to attract and retain workers (GiGroup, 2023).

Providing job security, competitive wages, appropriate remuneration, and meaningful work are key strategies in attracting and retaining talent across sectors (EVA, 2016).

Some port strategies are exploring how to improve their image through utilising role models, connecting port narratives to major societal change and enhancing visibility (SEOR, 2024).

In aviation, some companies have attempted the strategy of increasing salaries, and private bus transport from surrounding areas to the airport to facilitate commutes (ETF, 2023b). Similarly, a report on good practices in the warehousing sector identifies efficient transport connections to work through for example bike paths, bus shuttles or car sharing as a way to attract and retain labour force (Prologis, 2019). Some warehousing sector actors have been investing in more and higher quality amenities such as food, social areas, opportunities to play sports and child care to increase attractiveness (Prologis, 2019). UPS in the logistics area is offering flexible working arrangements and competitive compensation packages (Supply Chain Council of European Union, 2024).

Other initiatives in warehousing are focusing on community engagement and employee commitment (Prologis, 2019). For instance, logistics parks in the Netherlands use newsletters and narrowcasting screens to keep employees informed and foster a sense of community. Polish logistics parks offer internship programmes for students pursuing logistics degrees, providing hands-on experience and encouraging future talent to enter the sector.

Additionally, some researchers have highlighted the need for improved methods to reward worker productivity. Considering the relatively low base pay in this sector, financial incentives could further boost employee retention (Wandelt and Wang, 2024).

Promoting gender equality and diversity

Air traffic management stakeholders note gender equality is currently amongst the top priorities for the sector. Several European initiatives have been implemented to enhance the attractiveness of the aviation sector and promote gender equality. In France, the EU Aviation Safety Agency Gender Equality Action Plan works to increase the number of women in aviation by providing networking opportunities, promoting women in leadership, and offering mentorship programmes (ICAO, 2023). In Spain, the air navigation manager has a Gender Equality Plan that includes mentoring and training programmes aimed at increasing the representation of women in technical and leadership roles (ICAO, 2023). Lufthansa in Germany has introduced programmes to support work-life balance, offering flexible schedules and childcare facilities to promote gender inclusion (ICAO, 2023). Finally, the Austrian Airlines Diversity Strategy focuses on creating an inclusive environment through gender-neutral language, equal pay initiatives, and a robust framework for women's professional development (ICAO, 2023).

In one Italian municipality, the port sector addressed labour gaps through initiatives such as an international conference, a photo exhibition of Port of Livorno workers, a mural celebrating women's contributions, cultural events, and worker interviews (ETF, 2023e). According to women in the industry, good practices in day-to-day activities include ensuring access to well-fitting Personal Protective Equipment (PPE) for women, providing suitable accommodation for women on older vessels, and fostering a safe and inclusive working environment for all genders (Pike and Terpilowski, 2022). General good practices identified in a 2024 study on transport sector firms include the use of automated salary ranking tools to address pay discrepancies, quarterly IDE (Inclusion, Diversity, and Equity) reports to track investments and actions, and blind CV reviews to eliminate bias in recruitment (Intermobility, 2024). Additionally, recruitment processes based on equality principles, internal training on the benefits of inclusion and diversity, and setting quotas for women in senior management are further strategies aimed at improving workplace equity and representation (Intermobility, 2024). Companies have launched diversity and inclusion initiatives to further increase attractiveness to the sector. For example, DHL has a 'Women at DHL Global Forwarding, Freight' which enables more women to fill leadership roles and promote equal opportunities through carer support, transparency and work arrangements (DHL, 2021). Another example is XPO's implementation of KPIs at a European level to track their progress in gender inclusion, disability inclusion, and age diversity amongst other measures (XPO, 2025). The EU rail sector in partnership with the EU Commission has launched a 'Women in Rail Award 2024' to promote fairness, diversity and equality (European Commission, 2024a).

Stakeholders highlight that measures to promote gender equality not only protect female workers' rights but also help to tackle labour shortages and widen the pool of future air traffic controllers.

Automation

Automation can reduce employee workload and improve attractiveness by creating more meaningful jobs (Spencer, 2023). Faster and automated warehousing solutions that can do repetitive manual tasks that do not attract young people joining the workforce (Cobb, 2023). Automation is not new, but innovation in the area continues to increase efficiency. For example, Amazon extensively uses robotics in its warehouses to identify and store inventory 75% faster than previous methods (Dresser, 2023). For automation to succeed in improving working conditions, there should be a partnership approach, like in Germany, where automation has increased the chances of workers staying in employment (Spencer, 2023).

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